QUANTITY SURVEYING, CONTRACTS AND TENDERS

Unit 1: Introduction and Approximate Estimates

Q6. Prepare / Determine approximate estimated cost of building:

---

1 Feb 2023 Paper – Numerical

Q2 (c)

Prepare an approximate estimate of town hall building having plinth area equal to 1500 m^2. [5]

i) Plinth area rate = 3200 per m^2
ii) Special architectural treatment = 2% of cost of building
iii) Water supply and sanitary installation = 5% of cost of building
iv) Electric installation = 14% of cost of building
v) other services = 6% of cost of building
vi) Contingencies = 3% of overall cost of building
vii) Supervision charges = 8% of overall cost of building

---

2 March 2024 Paper – Numerical

Q2 (c)

Determine approximate estimated cost of a hospital building using following data. [5]

i) Number of beds in proposed Hospital = 135 Nos.
ii) Cost of construction for a hospital building of 125 beds and with same specification as the proposed hospital constructed 5 years before = Rs. 26,00,000/-
iii) Assume 20% rise in construction cost over rates before three years.
iv) Assume a provision of 12% of construction cost for water supply, drainage and electrification.

---

3 March 2025 Paper – Numerical

Q1 (b)

Determine approximate estimated cost of a residential building using following data. [5]

i) Built up area of the proposed building = 350 sq.m.

ii) Cost of construction for a building of built-up area 300 sq.m and with same specification as proposed building constructed 3 years before = Rs. 16,00,000/-

iii) Assume 24% rise in construction cost over rates before three years.

iv) Assume a provision of 15% of construction cost for water supply, drainage and electrification.

Q1. Explain administrative approval and technical sanction. ★

Q2. Enlist different types of approximate estimate and explain anyone. ★

Q3. Prepare standard format of measurement form and abstract form. ★

Q4. Explain valuation and its uses. State role of rate analysis in estimation. ★

Q5. Write short note on:

- Centage charges
- Lead statement ★

Q6. Prepare / Determine approximate estimated cost of building:

- Town hall (Plinth area method
- Hospital (Service unit method)
- Residential building (Cost index method) ★★

(Numerical repeated every year – very important)

Q7. Make a note on work charge establishment. ★

Q8. What is centage charge? ★

Q9. Enlist data required for preparing estimate (Explain in detail). ★★

Q10. Points considered while estimating irrigation type civil works. ★

Q11. State any five items with mode of measurement units. ★

Q12. Write short note on:

- Prime cost
- Provisional cost ★

Q13. Write short note on:

- Contingencies
- Rate analysis ★

Q14. Explain Bay method and Service unit method. ★

Unit-2: Tenders, Contracts and Arbitration

Q15. What is BOT and global tendering? ★

Q16. Explain P.W.D. procedure for execution of minor works. ★★

Q17. What is arbitration? Define and explain need of arbitration. ★★

Q18. What are powers of arbitrator? ★★

Q19. Discuss tendering procedure for specific construction work. ★

Q20. Explain:

- Defects liability period
- Liquidated damages ★★

Q21. Explain tender and give example. ★

Q22. Explain registration procedure for contractor. ★

Q23. State various methods of inviting tenders and explain anyone. ★

Q24. Explain five conditions of contract related to execution of work. ★

Q25. What is tender notice? Discuss essential contents. ★

Q26. Define contract and explain objectives. ★

Q27. Explain role of arbitrator in civil engineering works. ★

Q2 (c)

Prepare an approximate estimate of town hall building having plinth area equal to 1500 m². [5]

i) **Plinth area rate = 3200 per m²**
ii) **Special architectural treatment = 2% of cost of building**
iii) **Water supply and sanitary installation = 5% of cost of building**
iv) **Electric installation = 14% of cost of building**
v) **Other services = 6% of cost of building**
vi) **Contingencies = 3% of overall cost of building**
vii) **Supervision charges = 8% of overall cost of building**

Ans:

Approximate Estimate of Town Hall Building

(Plinth Area Method)

Given Data:

Plinth area = 1500 m²

Plinth area rate = Rs. 3200 per m²

Special architectural treatment = 2% of cost of building

Water supply & sanitary installation = 5% of cost of building

Electric installation = 14% of cost of building

Other services = 6% of cost of building

Contingencies = 3% of overall cost

Supervision charges = 8% of overall cost

Step 1: Calculate Building Cost

Building Cost = Plinth Area × Plinth Area Rate

= 1500 × 3200

= Rs. 48,00,000

Step 2: Add Special Architectural Treatment (2%)

2% of 48,00,000 = 96,000

Subtotal = 48,00,000 + 96,000
= Rs. 48,96,000

Step 3: Add Water Supply & Sanitary (5%)

5% of 48,00,000 = 2,40,000

Subtotal = 48,96,000 + 2,40,000
= Rs. 51,36,000

Step 4: Add Electric Installation (14%)

14% of 48,00,000 = 6,72,000

Subtotal = 51,36,000 + 6,72,000
= Rs. 58,08,000

Step 5: Add Other Services (6%)

6% of 48,00,000 = 2,88,000

Subtotal = 58,08,000 + 2,88,000
= Rs. 60,96,000

Step 6: Add Contingencies (3%)

3% of 60,96,000 = 1,82,880

Subtotal = 60,96,000 + 1,82,880
= Rs. 62,78,880

Step 7: Add Supervision Charges (8%)

8% of 62,78,880 = 5,02,310

Final Estimated Cost

Total Cost = 62,78,880 + 5,02,310
= Rs. 67,81,190

Final Answer:

Approximate Estimated Cost of Town Hall
= Rs. 67,81,190 (Approximately Rs. 67.8 Lakhs)

Q2 (c)

Determine approximate estimated cost of a hospital building using following data. [5]

- **i) Number of beds in proposed Hospital = 135 Nos.**
- **ii) Cost of construction for a hospital building of 125 beds and with same specification as the proposed hospital constructed 5 years before = Rs. 26,00,000/-**
- **iii) Assume 20% rise in construction cost over rates before three years.**
- **iv) Assume a provision of 12% of construction cost for water supply, drainage and electrification.**

Ans:

Approximate Estimate of Hospital Building

(Service Unit Method – Bed Method)

Given Data:

Number of beds in proposed hospital = 135 beds

Cost of construction of 125 beds hospital (5 years before)
= Rs. 26,00,000

Rise in construction cost = 20%

Provision for water supply, drainage & electrification = 12% of construction cost

---

Step 1: Find Cost Per Bed (Old Cost)

Cost per bed = Total Cost / Number of Beds

= 26,00,000 / 125

= Rs. 20,800 per bed

Step 2: Apply 20% Rise in Construction Cost

Increase = 20% of 20,800

= 0.20 × 20,800
= 4,160

New Cost per Bed = 20,800 + 4,160
= Rs. 24,960 per bed

Step 3: Calculate Construction Cost for 135 Beds

Construction Cost = 24,960 × 135

= Rs. 33,69,600

Step 4: Add 12% for Water Supply, Drainage & Electrification

12% of 33,69,600

= 0.12 × 33,69,600
= Rs. 4,04,352

Step 5: Calculate Total Estimated Cost

Total Cost = 33,69,600 + 4,04,352

= Rs. 37,73,952

Final Answer:

Approximate Estimated Cost of Proposed Hospital
= Rs. 37,73,952

(Approximately Rs. 37.74 Lakhs)

Q1 (b)

Determine approximate estimated cost of a residential building using following data. [5]

- **i) Built up area of the proposed building = 350 sq.m.**
- **ii) Cost of construction for a building of built-up area 300 sq.m and with same specification as proposed building constructed 3 years before = Rs. 16,00,000/-**
- **iii) Assume 24% rise in construction cost over rates before three years.**
- **iv) Assume a provision of 15% of construction cost for water supply, drainage and electrification.**

Approximate Estimate of Residential Building

(Cost Index Method / Area Method)

Given Data:

Built up area of proposed building = 350 sq.m

Cost of construction of 300 sq.m building (3 years before)
= Rs. 16,00,000

Rise in construction cost = 24%

Provision for water supply, drainage & electrification = 15% of construction cost

---

Step 1: Find Old Cost per sq.m

Cost per sq.m = Total Cost / Built up Area

= 16,00,000 / 300

= Rs. 5,333.33 per sq.m

Step 2: Apply 24% Rise in Construction Cost

Increase = 24% of 5,333.33

= 0.24 × 5,333.33
= Rs. 1,280

New Cost per sq.m = 5,333.33 + 1,280

= Rs. 6,613.33 per sq.m

Step 3: Calculate Construction Cost for 350 sq.m

Construction Cost = 6,613.33 × 350

= Rs. 23,14,665 (Approximately)

Step 4: Add 15% for Water Supply, Drainage & Electrification

15% of 23,14,665

= 0.15 × 23,14,665
= Rs. 3,47,200

Step 5: Calculate Total Estimated Cost

Total Cost = 23,14,665 + 3,47,200

= Rs. 26,61,865

Final Answer:

Approximate Estimated Cost of Proposed Residential Building
= Rs. 26,61,865

(Approximately Rs. 26.62 Lakhs)

Q1. Explain Administrative Approval and Technical Sanction. ★

Introduction

In government construction works, no project can start without proper approvals. Two important approvals required before starting any public work are:

• Administrative Approval
• Technical Sanction

Both approvals ensure that the work is necessary, financially possible, and technically correct.

Administrative Approval

Meaning

Administrative Approval is the formal approval given by the competent administrative authority to carry out a project.

It confirms that:

• The project is necessary
• Funds can be provided
• Work can be taken up

Purpose

• To approve the necessity of the project
• To approve the preliminary estimated cost
• To make budget provision
• To allow preparation of detailed drawings and estimates

Given By

• Government department

• Municipal authority

• Head of Department

• Higher administrative authority

Example

If a proposal is made to construct a government hospital costing Rs. 5 crores, the government first approves the need and budget for it. This approval is called Administrative Approval.

Technical Sanction

Meaning

Technical Sanction is the approval of detailed drawings, design, quantities and detailed estimate by the competent technical authority.

It confirms that:

• The design is structurally safe

• Quantities are correctly calculated

• Rates are properly applied

• The estimate is technically correct

Purpose

• To verify detailed drawings and specifications

• To check quantity calculations

• To ensure technical feasibility

• To control excess expenditure

Given By

• Executive Engineer

• Superintending Engineer

• Chief Engineer

(Depending upon cost limit)

Example

After administrative approval, detailed drawings and estimates of the hospital are prepared. The Executive Engineer checks and approves them technically. This is Technical Sanction.

Difference Between Administrative Approval and Technical Sanction

Administrative Approval	**Technical Sanction**
Given by administrative authority such as Government or Head of Department	Given by technical authority such as Executive Engineer or Chief Engineer
Approves the necessity of the project	Approves the technical correctness of the project
Based on preliminary estimate	Based on detailed estimate
Confirms availability of funds	Confirms correctness of quantities, rates and design
Comes before technical sanction	Comes after administrative approval
Concerned mainly with financial and administrative aspects	Concerned mainly with engineering and technical aspects

---

Conclusion

Administrative Approval ensures that the project is required and financially approved. Technical Sanction ensures that the project is technically correct and safe to execute.

Both approvals are compulsory before starting any government work.

Q2. Enlist Different Types of Approximate Estimate and Explain Any One. ★

Introduction

An approximate estimate is prepared to know the rough cost of a project before preparing a detailed estimate. It helps in getting administrative approval and budget provision.

It is also called a preliminary estimate or rough cost estimate.

Types of Approximate Estimate

• Plinth Area Method

• Cubical Content Method

• Service Unit Method

• Bay Method

• Approximate Quantity Method

• Cost Index Method

Explanation of Plinth Area Method

Meaning

In this method, the cost of building is calculated based on plinth area and plinth area rate.

Plinth area means the built-up covered area measured at floor level.

Formula

Approximate Cost = Plinth Area × Plinth Area Rate

Procedure

• Measure the plinth area of proposed building

• Find plinth area rate of similar type building

• Multiply area with rate

• Add percentage for services such as water supply, electrification, etc.

Example

If plinth area of building = 500 m^2
Plinth area rate = Rs. 3000 per m^2

Estimated Cost = 500×3000
= Rs. 15,00,000

Advantages

• Simple and quick method

• Requires less calculation

• Suitable for residential and public buildings

Limitations

• Not very accurate

• Does not consider height differences

• Depends on past data accuracy

Conclusion

Approximate estimate helps in knowing rough cost of project at initial stage. Among different methods, plinth area method is most commonly used because it is simple and quick.

Q3. Prepare Standard Format of Measurement Form and Abstract Form. ★

Introduction

In estimation and billing, quantities of work are recorded in a Measurement Form. After calculating total quantities and cost, they are summarized in an Abstract Form.

Both forms are essential in quantity surveying.

Measurement Form (Standard Format)

The measurement form is used to record detailed measurements of each item of work.

Standard Format of Measurement Form

Sr. No.	Description of Item	No.	Length (m)	Breadth (m)	Height/Depth (m)	Quantity	Remarks

Explanation of Columns

• Sr. No. – Serial number of items

• Description of Item – Name of work item

• No. – Number of units

• Length – Measured length

• Breadth – Measured width

• Height/Depth – Measured height or depth

• Quantity – Calculated quantity

• Remarks – Any additional note

Abstract Form (Standard Format)

Abstract form is used to summarize total quantity and cost of each item.

Standard Format of Abstract Form

Sr. No.	Description of Item	Quantity	Unit	Rate (Rs.)	Amount (Rs.)

Explanation of Columns

• Sr. No. – Serial number

• Description of Item – Name of work

• Quantity – Total quantity from measurement sheet

- Unit – Unit of measurement (m^3, m^2, m, No.)
- Rate – Rate per unit
- Amount – Quantity × Rate

Importance of Measurement and Abstract Forms

- Helps in accurate quantity calculation
- Useful for preparation of estimate
- Required for billing and payment
- Maintains proper record of work

Conclusion

Measurement form records detailed dimensions of work items, while abstract form summarizes quantities and cost. Both are essential documents in quantity surveying and preparation of estimates.

Q4. Explain Valuation and Its Uses. State Role of Rate Analysis in Estimation. ★

Introduction

In civil engineering, valuation is the process of determining the present market value of a property such as land, building, or structure.

It is important for financial, legal and taxation purposes.

Rate analysis is used to determine the cost per unit of an item of work. It helps in preparing accurate estimates.

Valuation

Meaning

Valuation is the process of estimating the present value of a property based on its location, condition, income and market demand.

Uses of Valuation

- To determine selling price of property
- For purchasing or leasing property
- For obtaining bank loan (mortgage purpose)
- For taxation purpose
- For insurance purpose
- For compulsory acquisition by government
- For fixation of rent
- For partition of property

Role of Rate Analysis in Estimation

Meaning of Rate Analysis

Rate analysis is the process of determining the cost per unit of an item of work.

It includes cost of:

- Materials
- Labou
- Equipment
- Transportation
- Overheads
- Contractor’s profit

Importance of Rate Analysis

- Helps in preparing detailed estimate
- Ensures correct and reasonable rates
- Avoids overestimation and underestimation
- Useful in tender preparation

- Helps in cost control
- Required for checking contractor’s rates

Example

If rate of brick masonry per cubic meter is required, rate analysis will calculate cost of:

- Bricks
- Cement and sand
- Labour charges
- Water charges
- Contractor’s profit

Total of all these gives rate per cubic meter.

Conclusion

Valuation determines the present market value of property for financial and legal purposes. Rate analysis plays an important role in estimation by determining correct unit rates and ensuring accuracy in project costing.

Q5. Write Short Note on:

- **Centage Charges**
- **Lead Statement ★**

Centage Charges

Meaning

Centage charges are the percentage charges levied by a government department or agency for executing a work on behalf of another department or organization.

It is charged for providing services such as supervision, planning, and execution of work.

Purpose of Centage Charges

- To cover supervision expenses
- To cover establishment expenses
- To meet administrative costs
- To cover planning and execution cost

Percentage of Centage Charges

Generally, ranges between 5% to 15% of the estimated cost depending on department rules.

Example

If a Municipal Corporation assigns construction of a building to PWD and the project cost is Rs. 1 crore, and centage charge is 10%, then:

Centage charge = 10% of 1 crore
= Rs. 10 lakhs

Lead Statement

Meaning

Lead statement shows the distance between the source of material and the construction site.

It is prepared to calculate transportation cost of materials.

Purpose of Lead Statement

- To calculate cost of carriage of materials
- To determine transportation charges
- To prepare accurate rate analysis
- To avoid excess payment for transport

Example

If sand is brought from a quarry located 12 km away from site, then lead = 12 km.

Transportation cost is calculated based on this lead distance.

Conclusion

Centage charges are percentage charges collected for execution and supervision of work. Lead statement helps in calculating transportation cost of materials. Both are important in preparation of estimates and rate analysis.

Q7. Make a Note on Work Charged Establishment. ★

Introduction

In government construction works, some staff are appointed temporarily for a specific project. These employees are called work charged establishment staff.

Their salary is charged directly to the cost of that particular work.

Meaning

Work charged establishment refers to temporary staff appointed for execution of a specific work and whose wages are included in the project cost.

Their employment continues only till the completion of the work.

Features of Work Charged Establishment

- Staff is temporary in nature
- Appointed only for specific project
- Salary is charged to the work estimate
- Service ends after completion of work
- Not part of permanent establishment

Examples of Work Charged Staff

- Work assistants
- Site supervisors
- Junior engineers
- Clerks
- Watchmen
- Storekeepers

Purpose

- To supervise site activities
- To maintain site records
- To assist in execution of work
- To ensure smooth progress of construction

Advantages

- Reduces burden on permanent staff
- No long-term financial liability
- Suitable for short-term projects

Conclusion

Work charged establishment includes temporary staff appointed for a specific project and paid from project funds. It helps in proper supervision and execution of government works without increasing permanent staff strength.

Q8. What is Centage Charge? ★

Introduction

In government construction works, sometimes one department executes work on behalf of another department. For providing this service, a certain percentage of the project cost is charged. This is known as centage charge.

Meaning

Centage charge is a percentage amount charged by a government department or agency for executing, supervising and managing a work on behalf of another department or organization.

It is added to the estimated cost of the project.

Purpose of Centage Charge

- To cover supervision expenses
- To cover establishment expenses
- To meet administrative costs
- To cover planning and execution expenses
- To compensate the executing department

Percentage of Centage Charge

- Generally, varies from 5% to 15% of the estimated cost
- Depends on departmental rules and type of work

Example

If the estimated cost of building is Rs. 50 lakhs and centage charge is 10%, then

Centage charge = 10% of 50,00,000

= Rs. 5,00,000

Total project cost = 55,00,000

Conclusion

Centage charge is a percentage amount added to the project cost when one department executes work for another. It helps cover supervision, administrative and establishment expenses of the executing agency.

Q9. Enlist Data Required for Preparing Estimate (Explain in Detail). ★★

Introduction

Before preparing any detailed estimate, proper data collection is necessary. An accurate estimate depends on correct and complete data.

If data is incomplete, the estimate may be wrong and lead to financial loss.

Data Required for Preparing Estimate

Drawings

- Plan
- Elevation
- Section
- Detailed drawings

Drawings help in calculating quantities of various items of work.

Specifications

- General specifications
- Detailed specifications

Specifications describe quality of materials, workmanship and method of construction.

They help in selecting proper rates.

Schedule of Rates (SOR)

- Government approved rates
- Standard rates for materials and labour

Used for calculating cost of each item.

Current Market Rates

- Latest cost of materials
- Labour wages
- Equipment charges

Required when SOR is not available or outdated.

Site Information

- Location of site
- Type of soil
- Availability of materials
- Water and electricity availability
- Transportation facilities

These factors affect overall cost.

Lead and Lift Details

- Distance from source of material
- Height or depth of work

Used to calculate transportation cost and additional charges.

Standard Data Book

- Provides material consumption
- Labour requirement
- Productivity rates

- Useful for rate analysis.

Rate Analysis

- Breakdown of material cost
- Labour cost
- Equipment cost
- Contractor's profit

Helps in fixing accurate unit rates.

Contingencies and Other Charges

- Provision for unforeseen expenses
- Supervision charges
- Centage charges
- Work charged establishment

Importance of Proper Data

- Ensures accurate estimate
- Avoids cost overrun
- Helps in budget planning
- Useful for tender preparation
- Prevents financial loss

Conclusion

Preparation of estimate requires detailed drawings, specifications, rates, site information and standard data. Proper data collection ensures accuracy, reliability and financial control of the project.

Q10. Points Considered While Estimating Irrigation Type Civil Works. ★

Introduction

Irrigation works include canals, dams, weirs, barrages, distributaries and related hydraulic structures.

While preparing estimate for irrigation works, special technical and site-related factors must be considered because such projects involve large earthwork and water flow.

Points to be Considered

Site Investigation

- Type of soil
- Ground water level
- Nature of foundation
- Topography of area

These factors affect excavation and foundation cost.

Type of Structure

- Canal
- Dam
- Weir
- Barrage
- Aqueduct

Different structures require different estimation approach.

Earthwork Calculation

- Large quantity of excavation
- Embankment work
- Cutting and filling
- Shrinkage and compaction allowance

Earthwork forms major part of irrigation project cost.

Lead and Lift

- Distance of earth or material transport
- Height of lifting material

Transportation cost significantly affects estimate.

Availability of Materials

- Source of sand
- Source of stone
- Availability of cement
- Transportation facilities

Material availability affects rate analysis.

Water Control Arrangements

- Temporary diversion channels
- Dewatering arrangements
- Cofferdams

These increase construction cost.

Hydraulic Design Considerations

- Discharge capacity
- Slope of canal
- Lining requirement
- Seepage control

Design parameters influence quantity of work.

Structures in Canal

- Cross drainage works
- Culverts
- Regulators
- Head works

These additional structures increase total project cost.

Machinery and Equipment

- Excavators
- Pumps
- Concrete mixers

Large projects require heavy equipment, increasing cost.

Contingencies and Supervision

- Provision for unforeseen work
- Work charged establishment
- Supervision charges

Conclusion

While estimating irrigation works, careful consideration of soil conditions, earthwork, lead and lift, hydraulic design and material availability is necessary. Proper evaluation of these factors ensures accurate and reliable cost estimation.

Q11. State Any Five Items with Mode of Measurement Units. ★

Introduction

In estimation and quantity surveying, each item of work is measured in a specific unit as per standard rules. The mode of measurement depends on the nature of work such as volume, area, length or number.

Five Items with Mode of Measurement

Earthwork in Excavation

- Measured in cubic meter (m^3)
- Mode of measurement – Length × Breadth × Depth

Plain Cement Concrete (PCC)

- Measured in cubic meter (m^3)
- Mode of measurement – Length × Breadth × Thickness

Brick Masonry

- Measured in cubic meter (m^3)
- Mode of measurement – Length × Breadth × Height

Plastering

- Measured in square meter (m^2)
- Mode of measurement – Length × Height

Reinforcement Steel

- Measured in kilogram (kg) or tone (t)
- Mode of measurement – Based on weight of steel used

Flooring

- Measured in square meter (m^2)
- Mode of measurement – Length × Breadth

Doors and Windows

- Measured in number (No.) or square meter (m^2)
- Mode of measurement – As per design size

Conclusion

Different construction items are measured in cubic meter, square meter, running meter, kilogram or number depending on the type of work. Proper mode of measurement ensures accurate quantity calculation and correct estimation.

Q12. Write Short Note on:

- **Prime Cost**
- **Provisional Cost ★**

Prime Cost

Meaning

Prime Cost (PC) is the estimated amount included in the contract for supply of specific materials or items whose exact price is not known at the time of preparing estimate.

It usually covers the cost of material only, excluding installation charges.

Features

- Included in estimate as a fixed allowance
- Used for special items such as sanitary fittings, tiles, electrical fittings
- Final payment is made based on actual purchase cost
- Contractor is paid actual cost-plus profit and overheads

Example

If high-quality sanitary fittings are to be installed but their exact brand is not finalized, a prime cost of Rs. 1,00,000 may be included in estimate.

After purchase, actual cost is adjusted in final bill.

Provisional Cost

Meaning

Provisional Cost (or Provisional Sum) is an estimated amount included in the contract for work which cannot be completely defined or measured at the time of preparing estimate.

It includes cost of both materials and labour.

Features

- Used when details of work are not fully known
- Included as a temporary amount
- Adjusted after actual work is completed
- Covers uncertain or future work

Example

Landscaping work may not be fully designed during estimation. A provisional sum of Rs. 2,00,000 may be included. After execution, actual cost is calculated and adjusted.

Difference Between Prime Cost and Provisional Cost

Prime Cost	**Provisional Cost**
For supply of specific materials	For complete work item

Covers material cost only	Covers material and labour cost
Used when item is selected later	Used when work details are not clear
Adjusted based on actual purchase	Adjusted based on actual execution

---

Conclusion

Prime cost is an allowance for supply of materials whose exact price is not known, while provisional cost is an estimated amount for work not fully defined at the time of estimation. Both help manage uncertainty in construction contracts.

Q13. Write Short Note on:

- **Contingencies**
- **Rate Analysis ★**

Contingencies

Meaning

Contingencies are the percentage amount added to the estimated cost to cover unforeseen expenses during execution of work.

These expenses cannot be predicted accurately at the time of preparing estimate.

Purpose of Contingencies

- To cover unexpected minor changes
- To meet unforeseen site conditions
- To cover small omissions in estimate
- To handle price fluctuations

Percentage of Contingencies

- Generally, 3% to 5% of estimated cost
- Depends on department rules

Example

If estimated cost of building = Rs. 50,00,000

Contingencies at 3%

Contingency amount = Rs. 1,50,000

Rate Analysis

Meaning

Rate analysis is the process of determining the cost per unit of an item of work.

It includes detailed calculation of material, labour, equipment and other charges.

Components of Rate Analysis

- Cost of materials
- Cost of labour
- Transportation charges
- Water and electricity charges
- Overheads
- Contractor’s profit

Importance of Rate Analysis

- Helps in preparing accurate estimate
- Ensures reasonable rates
- Useful for tender preparation
- Helps in cost control
- Required to check contractor’s rates

Example

For brick masonry per cubic meter, rate analysis includes cost of:

- Bricks
- Cement and sand
- Labour charges
- Contractor’s profit

Conclusion

Contingencies are added to meet unforeseen expenses, while rate analysis helps determine correct unit rates of work items. Both are essential for accurate estimation and cost control in construction projects.

Q14. Explain Bay Method and Service Unit Method. ★

Introduction

Bay method and Service unit method are types of approximate estimation methods. These methods are used to calculate rough cost of a project at the preliminary stage.

They are useful for getting administrative approval and budget provision.

Bay Method

Meaning

Bay method is used for buildings that consist of several similar bays.

A bay is the space between two consecutive columns or frames.

Procedure

- Calculate the cost of one typical bay
- Count total number of bays in building
- Multiply cost per bay by total number of bays
- Add cost of special features if required

Suitable For

- Factory buildings
- Warehouses
- Industrial sheds
- Workshop buildings

Example

If cost of one bay = Rs. 2,00,000

Total number of bays = 10

Estimated Cost = 2,00,000 × 10

= Rs. 20,00,000

Service Unit Method

Meaning

Service unit method is used for buildings where cost depends on number of service units.

Service unit may be bed, seat, student, room, etc.

Procedure

- Find cost per service unit from similar building
- Multiply cost per unit by number of units required
- Add percentage for services such as water supply and electrification

---

Suitable For

- Hospitals (cost per bed)
- Schools (cost per student)
- Hostels (cost per room)
- Hotels (cost per room)

Example

If cost per hospital bed = Rs. 25,000

Number of beds = 100

Estimated Cost = 25,000 × 100

= Rs. 25,00,000

Conclusion

Bay method is suitable for buildings with repeated structural units, while service unit method is suitable for buildings where cost depends on number of functional units. Both methods are simple and useful for approximate estimation.

Q15. What is BOT and Global Tendering? ★

Introduction

In large infrastructure projects, different methods are used for financing and execution of work. BOT and Global Tendering are two such important concepts used in construction and public works.

BOT (Build – Operate – Transfer)

Meaning

BOT is a project delivery method in which a private company builds a project, operates it for a fixed period to recover investment, and then transfers it to the government.

Process of BOT

- Private company finances the project
- Company constructs the project
- Company operates and maintains the project for a concession period
- Investment is recovered through tolls, fees or charges
- After completion of concession period, project is transferred to government

Features

- No immediate financial burden on government
- Private sector participation
- Risk is shared between government and private party
- Suitable for large infrastructure projects

Examples

- Highways and expressways
- Bridges
- Airports
- Power plants

Global Tendering

Meaning

Global tendering is a method of inviting tenders from international contractors or companies.

It allows participation of foreign firms in large or specialized projects.

Features

- Open to international bidders
- Used for large-scale or technically complex projects
- Encourages competition
- Ensures advanced technology and expertise

Purpose

- To get best technology
- To get competitive rates
- To improve quality of work
- To attract foreign investment

Example

Construction of metro rail system may invite global tenders so that foreign companies with advanced technology can participate.

Conclusion

BOT is a project execution method involving private investment and later transfer to government, while global tendering is a method of inviting bids from international companies. Both are important in modern infrastructure development.

Q16. Explain P.W.D. Procedure for Execution of Minor Works. ★★

Introduction

Public Works Department (PWD) follows a systematic procedure for execution of minor works such as small buildings, repairs, culverts, and maintenance works.

Minor works are generally low-cost works within the financial powers of local authorities.

P.W.D. Procedure for Execution of Minor Works

Preparation of Estimate

- Site inspection is carried out
- Rough estimate is prepared
- Detailed estimate with quantities and rates is prepared
- Estimate is checked and approved

Administrative Approval

- Approval of necessity of work is obtained
- Budget provision is confirmed
- Work is officially sanctioned

Technical Sanction

- Detailed drawings and estimate are verified
- Quantities and rates are checked

- Technical approval is given by competent engineer

Tendering Process

- Tender notice is prepared
- Tenders are invited
- Received tenders are opened and compared
- Lowest suitable bidder is selected

Issue of Work Order

- Work order is issued to contractor
- Agreement is signed
- Time limit for completion is mentioned

Execution of Work

- Site is handed over to contractor
- Work is executed as per drawings and specifications
- Supervision is done by PWD staff
- Quality control is maintained

Measurement of Work

- Measurements are recorded in Measurement Book
- Quantities are verified
- Running bills are prepared

Payment of Bills

- Bills are checked and verified
- Payment is made as per work done
- Final bill is prepared after completion

Completion and Inspection

- Work is inspected by engineer
- Defects are rectified
- Completion certificate is issued

Conclusion

PWD follows a proper procedure including estimation, approval, tendering, execution and payment for minor works. This ensures transparency, quality control and proper use of public funds.

Q17. What is Arbitration? Define and Explain Need of Arbitration. ★★

Introduction

In construction projects, disputes may arise between the contractor and the department regarding payment, delay, quality of work or contract conditions.

To settle such disputes without going to court, arbitration is used.

Meaning of Arbitration

Arbitration is a method of resolving disputes outside the court by appointing a neutral third person called an arbitrator.

The arbitrator hears both parties, examines evidence and gives a decision known as an award.

Definition

Arbitration is a legal process in which disputes between two parties are settled by a mutually agreed third person instead of a court of law.

Need of Arbitration

Quick Settlement

- Court cases take many years
- Arbitration provides faster resolution

Cost Effective

- Court litigation is expensive
- Arbitration reduces legal expenses

Technical Expertise

- Construction disputes involve technical matters
- Arbitrator may be an engineer or technical expert

Confidential Process

- Proceedings are private
- Protects business reputation

Binding Decision

- Decision of arbitrator is legally binding
- Reduces further disputes

Maintains Business Relationship

- Less formal and less hostile than court
- Helps maintain professional relations

Example

If a contractor claims extra payment for additional work but department refuses, the matter can be referred to arbitration instead of filing a court case.

Conclusion

Arbitration is an effective method of dispute resolution in civil engineering contracts. It ensures quick, economical and fair settlement of disputes without lengthy court procedures.

Q18. What are Powers of Arbitrator? ★★

Introduction

An arbitrator is a neutral person appointed to resolve disputes between two parties in a contract.

In construction works, the arbitrator has certain legal powers to ensure fair settlement of disputes.

Powers of Arbitrator

Power to Hear Both Parties

- Arbitrator can call both parties for hearing
- Can allow them to present arguments and evidence

Power to Call for Documents

- Can demand relevant documents such as contract agreement, drawings, bills, correspondence
- Can examine records like Measurement Book and site reports

Power to Take Evidence

- Can examine witnesses
- Can take written or oral statements
- Can record technical evidence

Power to Inspect Site

- Arbitrator can visit the construction site
- Can verify actual work done
- Helps in understanding technical disputes

Power to Appoint Experts

- Can appoint technical experts if required
- Experts may give opinion on quality, delay, extra items etc.

Power to Decide Disputes

- Can decide claims related to payment, delay, extra work, compensation
- Decision given is called "Arbitral Award"

Power to Grant Relief

- Can award money compensation
- Can allow extension of time
- Can impose interest on delayed payments

Power to Make Final Award

- Arbitrator gives final written decision
- Award is binding on both parties
- It can be enforced legally

Conclusion

The arbitrator has sufficient legal powers to investigate, examine and settle disputes fairly. His decision helps avoid long court cases and ensures speedy justice in civil engineering contracts.

Q19. Discuss Tendering Procedure for Specific Construction Work. ★

Introduction

Tendering is the process of inviting contractors to submit their rates for a construction work.

It ensures transparency, competition and selection of a suitable contractor.

For explanation, consider construction of a **residential building** as an example.

Tendering Procedure

Preparation of Detailed Estimate

- Detailed drawings and specifications are prepared
- Quantities are calculated
- Estimated cost is worked out
- Technical sanction is obtained

Preparation of Tender Documents

Tender documents generally include:

- Notice Inviting Tender (NIT)
- General conditions of contract
- Special conditions
- Specifications
- Bill of Quantities (BOQ)
- Form of agreement

Invitation of Tender

- Tender notice is published in newspaper or online portal
- Contractors are invited to submit bids
- Last date and time of submission is mentioned

Submission of Tender

- Contractors submit sealed tenders or online bids
- Earnest Money Deposit (EMD) is attached
- Required documents are submitted

Opening of Tender

- Tenders are opened on specified date
- Rates quoted by contractors are read out
- Comparative statement is prepared

Evaluation of Tender

- Lowest bidder (L1) is identified
- Technical qualifications are checked
- Financial capacity and experience are verified

Acceptance of Tender

- Tender is approved by competent authority
- Letter of Acceptance is issued
- Contractor deposits security deposit

Signing of Agreement

- Formal contract agreement is signed

- Work order is issued
- Time for completion is mentioned

Example

For construction of a residential building worth ₹50 lakh:

- Estimate is prepared
- Tenders are invited
- Lowest qualified contractor is selected
- Agreement is signed and work starts

Conclusion

Tendering procedure ensures fair competition, transparency and proper selection of contractor. It protects public funds and ensures quality construction work.

Q20. Explain:

- **Defects Liability Period**
- **Liquidated Damages ★★**

Introduction

In construction contracts, certain clauses are included to protect the owner against poor quality work and delay in completion.

Two important terms are Defects Liability Period and Liquidated Damages.

Defects Liability Period (DLP)

Meaning

Defects Liability Period is the period after completion of work during which the contractor is responsible for correcting defects at his own cost.

It usually starts from the date of completion certificate.

Features

- Generally, 6 months to 12 months
- Contractor must repair cracks, leakage, faulty plaster, etc.
- No extra payment is made for rectification
- Security deposit may be released after DLP

Example

If cracks appear in walls within 6 months after completion, the contractor must repair them free of cost.

Liquidated Damages (LD)

Meaning

Liquidated damages are the fixed amount of money that contractor must pay to the owner if work is not completed within the specified time.

It is a penalty for delay.

Features

- Amount is mentioned in contract
- Usually calculated per day or per week of delay
- Deducted from contractor's payment
- Maximum limit is specified

Example

If contract states ₹5,000 per day as LD and work is delayed by 10 days:

Penalty = ₹5,000 × 10

Penalty = ₹50,000

This amount is deducted from contractor's bill.

Difference Between DLP and LD

Defects Liability Period	**Liquidated Damages**
Related to quality of work	Related to delay in time
Starts after completion	Applies before completion
Contractor repairs defects	Contractor pays penalty
No extra payment given	Amount deducted from bill

Conclusion

Defects Liability Period ensures quality of work after completion, while Liquidated Damages ensure timely completion of work. Both clauses protect the owner's interest in civil engineering contracts.

Q21. Explain Tender and Give Example. ★

Introduction

In construction projects, work is usually not given directly to a contractor.

Instead, contractors are invited to submit their rates. This process is called tendering, and the offer submitted by contractor is called a tender.

Meaning of Tender

A tender is a written offer submitted by a contractor to execute a specific work at specified rates within a specified time under certain conditions.

It is a formal proposal to carry out construction work.

Definition

Tender is an offer made by a contractor in response to a tender notice to execute a work according to given drawings, specifications and conditions at quoted rates.

Important Features of Tender

- It is a written document
- Submitted before last date and time
- Includes quoted rates
- Includes Earnest Money Deposit (EMD)
- Based on drawings and specifications
- Legally binding after acceptance

Documents Included in Tender

- Notice Inviting Tender
- General conditions of contract
- Special conditions
- Bill of Quantities (BOQ)
- Form of agreement
- Specifications and drawings

Example

Suppose Public Works Department invites tender for construction of a school building costing ₹80 lakh.

- Three contractors submit their tenders

- Contractor A quotes ₹78 lakh
- Contractor B quotes ₹82 lakh
- Contractor C quotes ₹76 lakh

After evaluation, the lowest suitable bidder (Contractor C) is selected.

The tender of Contractor C is accepted and work order is issued.

Importance of Tender

- Ensures competition
- Provides transparency
- Helps in selecting qualified contractor
- Protects public funds

Conclusion

Tender is a formal written offer by a contractor to execute construction work at specified rates and conditions. It ensures fairness, transparency and proper selection of contractor in civil engineering projects.

Q22. Explain Registration Procedure for Contractor. ★

Introduction

Before participating in government tenders, a contractor must register with the concerned department such as PWD.

Registration ensures that only qualified and financially capable contractors are allowed to execute public works.

Purpose of Registration

- To verify technical qualification

- To check financial capacity
- To ensure experience in similar works
- To maintain quality and transparency

Registration Procedure

Submission of Application

- Contractor submits prescribed application form
- Application is submitted to concerned department

Submission of Required Documents

Documents generally required:

- Educational qualification certificate
- Experience certificate of completed works
- Financial statements
- Income tax clearance certificate
- GST registration
- PAN card
- Bank solvency certificate

Payment of Registration Fees

- Registration fee is paid as per category
- Fees vary depending on class of contractor

Verification of Documents

- Department verifies technical and financial capacity
- Experience of similar work is checked
- Past performance is examined

Classification of Contractor

Contractors are classified into categories such as:

- Class I – For high value works
- Class II – Medium value works
- Class III – Small value works

Registration limit depends on class.

Issue of Registration Certificate

- After approval, registration certificate is issued
- Valid for specific period
- Contractor can participate in tenders within prescribed limit

---

Example

If a contractor wants to execute works up to ₹50 lakh, he must register under suitable class by submitting documents and paying registration fees.

Conclusion

Registration of contractor is essential for maintaining quality, eligibility and accountability in public construction works. It ensures that only capable contractors are allowed to execute government projects.

Q23. State Various Methods of Inviting Tenders and Explain Any One. ★

Introduction

Inviting tenders is an important step in construction projects.

Different methods are used depending on nature, cost and urgency of work.

Various Methods of Inviting Tenders

- Open Tender
- Limited Tender
- Selective Tender
- Negotiated Tender
- Global Tender
- E-Tendering

Explanation of Open Tender

Meaning

Open tender is a method in which tender notice is published publicly and any eligible contractor can participate.

Features

- Tender notice published in newspaper or online portal
- Open to all registered contractors
- Ensures maximum competition
- Transparent method

Procedure

- Notice Inviting Tender is issued
- Contractors purchase tender documents
- Tenders are submitted before last date
- Tenders are opened and evaluated
- Lowest suitable bidder is selected

Advantages

- Fair and transparent
- Competitive rates
- Equal opportunity to all contractors

Example

For construction of a government hospital building, PWD publishes tender notice in newspaper.

All eligible contractors can submit their bids.

Conclusion

Different methods of inviting tenders are used based on project requirements. Open tender is the most common and transparent method in public construction works.

Q24. Explain Five Conditions of Contract Related to Execution of Work. ★

Introduction

A contract contains various conditions to ensure proper execution of construction work.

These conditions define rights, duties and responsibilities of both contractor and owner.

Important Conditions Related to Execution of Work

Time of Completion

- Contractor must complete work within specified time
- Time schedule is mentioned in agreement
- Delay may attract penalty

Example: Work must be completed within 12 months.

Quality of Materials and Workmanship

- Materials must be as per specifications
- Engineer has right to reject poor quality work
- Contractor must follow standard codes

Example: Cement must be of approved brand.

Payment Terms

- Payment is made as per measured work
- Running bills are paid periodically
- Final bill is paid after completion

Example: Payment made every month based on progress.

Security Deposit

- Contractor deposits certain percentage of contract value
- Acts as security for proper performance
- Released after defects liability period

Example: Five percent of contract value kept as security.

Extra Items and Variations

- Extra work must be approved in writing
- Rates are decided as per contract
- Payment made accordingly

Example: Additional foundation depth due to weak soil.

Conclusion

Conditions of contract ensure timely completion, quality work and fair payment. They protect interests of both contractor and owner and help in smooth execution of civil engineering projects.

Q25. What is Tender Notice? Discuss Essential Contents. ★

Introduction

Before starting any construction work, the department invites contractors to submit their bids.

This invitation is given through a document called a Tender Notice.

Meaning of Tender Notice

Tender Notice is a public announcement issued by the department inviting contractors to submit tenders for a specific work.

It is also called Notice Inviting Tender (NIT).

Purpose of Tender Notice

- To inform contractors about the proposed work
- To ensure transparency
- To invite competitive bids
- To select suitable contractor

Essential Contents of Tender Notice

Name of Department

- Name of PWD or concerned authority

Name and Description of Work

- Nature of work
- Location of project

Example: Construction of school building at Pune.

Estimated Cost

- Approximate cost of work
- Helps contractors decide eligibility

Earnest Money Deposit (EMD)

- Amount of EMD required
- Mode of payment

Cost of Tender Form

- Price of tender documents
- Non-refundable amount

Time of Completion

- Time period within which work must be completed

Eligibility Criteria

- Required class of contractor
- Experience requirements

Date and Time

- Last date for submission
- Date and time of opening tender

Contact Details

- Office address
- Phone or email for enquiry

Example

PWD issues tender notice for construction of a hospital costing ₹2 crore.

The notice includes estimated cost, EMD of ₹2 lakh, completion time of 18 months and last date for submission.

Conclusion

Tender notice is an important document that initiates the tendering process. It provides all necessary information to contractors and ensures transparency and fair competition in civil engineering works.

Q26. Define Contract and Explain Objectives. ★

Introduction

In civil engineering works, an agreement between owner and contractor is necessary before starting construction.

This agreement is called a contract.

Meaning of Contract

A contract is a legally binding agreement between two or more parties to perform certain work under specified terms and conditions.

In construction, it is an agreement between the owner (department) and the contractor.

Definition

A contract is an agreement enforceable by law, in which one party agrees to execute work and the other party agrees to pay for it under specified conditions.

Essential Elements of Contract

- Offer
- Acceptance
- Lawful consideration
- Free consent
- Competent parties
- Lawful object

Objectives of Contract

Clear Understanding

- Defines scope of work
- Specifies responsibilities of both parties

Legal Protection

- Provides legal security
- Protects rights of owner and contractor

Time and Cost Control

- Specifies contract amount
- Mentions time of completion
- Avoids unnecessary disputes

Quality Assurance

- Specifies materials and workmanship
- Ensures work as per standards

Dispute Resolution

- Includes clauses for arbitration
- Provides method to settle disputes

Financial Security

- Includes security deposit
- Specifies payment terms

Example

If PWD awards contract of ₹1 crore for construction of bridge, the agreement will mention:

- Scope of work
- Completion period
- Payment terms
- Penalty for delay
- Quality standards

Conclusion

A contract is a legally binding agreement that ensures smooth execution of construction work. It defines duties, rights and responsibilities of both parties and helps avoid misunderstandings and disputes.

Q27. Explain Role of Arbitrator in Civil Engineering Works. ★

Introduction

In civil engineering contracts, disputes may arise between the contractor and the owner regarding payment, delay, quality of work, extra items or interpretation of contract clauses.

To resolve such disputes without going to court, an arbitrator is appointed.

Meaning of Arbitrator

An arbitrator is a neutral and independent person appointed to settle disputes between two parties in a contract.

He acts as a judge outside the court.

Role of Arbitrator in Civil Engineering Works

Settlement of Disputes

- Resolves disputes related to payment, delay, extra work and compensation
- Gives fair and impartial decision

Examination of Contract

- Studies contract agreement
- Checks terms and conditions

- Interprets clauses properly

Hearing Both Parties

- Provides equal opportunity to both parties
- Listens to arguments and evidence
- Ensures natural justice

Verification of Records

- Examines documents such as bills, drawings and Measurement Book
- Reviews correspondence and reports

Site Inspection

- May visit construction site
- Verifies actual work executed
- Understands technical issues

Decision Making

- Gives written decision called “Arbitral Award”
- Decides amount payable, compensation or extension of time
- Award is legally binding

Maintaining Confidentiality

- Arbitration proceedings are private
- Protects reputation of both parties

Example

If contractor claims extra payment for additional excavation and department refuses, the dispute is referred to arbitrator.

After examining records and hearing both parties, the arbitrator gives final decision.

Conclusion

The arbitrator plays an important role in civil engineering works by resolving disputes quickly and fairly. He helps avoid lengthy court cases and ensures smooth completion of construction projects.
